University of Bozen-Bolzano
Bachelor's Degree in Informatics and Management of Digital Business

Economics of Digital Markets

Problem set #2

Academic Year 2026/27

Instructor: Nicola Campigotto

The data

In this assignment, you will analyze experimental data from a public goods game without punishment. The experiment was conducted in class at the beginning of the first lecture. Participants were randomly assigned to groups of four anonymous players and played the same game for 10 rounds. Group members remained unchanged throughout the experiment.

At the beginning of each round, each player received an endowment of 20 points and decided how many points to contribute to a public project. The group's total contribution was multiplied by 1.6, and the resulting amount was divided equally among its four members. Therefore, each player earned 0.4 points for every point contributed to the project by any group member.

At the end of each round, players were informed of the group's total contribution, each member's individual contribution, and their own payoff. A player's payoff in a given round was the sum of the points they kept and the points they received from the public project.

The dataset is provided as a `.csv` file and contains the following variables.

- `player_id`: Unique player identifier
- `round`: Round identifier (10 rounds in total)
- `group`: Unique group identifier
- `group_member`: Player number within the group (each group has four members)
- `contribution_round`: Number of points the player contributed to the project in the current round
- `payoff_round`: Player's payoff in the current round
- `cumulative_payoff`: Total payoff earned up to and including the current round

Use any software you prefer to analyse the data and answer the following questions.

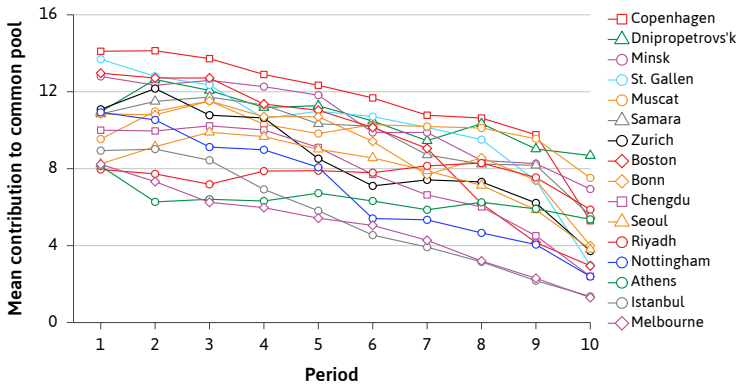
Questions

1. Plot the average contribution over time for the full sample of observations. What pattern do you observe? How does it compare with the results reported by Hermann et al. (2008), shown in Figure 1?

2. Plot the average contribution over time for each group. Do the four groups show similar patterns, or are there clear differences? Do any groups behave as predicted by the standard theory of the public goods game? Do any groups clearly depart from that prediction?

3. Focus on groups 2 and 3. Create two separate plots, one for each group, showing each group member's contributions over time. Interpret the behaviour of the members of each group and discuss the possible reasoning behind their decisions.

Figure 1: *The public goods game around the world (source: Hermann et al. 2008)*



References

Hermann, Benedikt, Christian Thöni, and Simon Gächter (2008), "Antisocial punishment across societies", *Science*, vol. 319, no. 5868, pp. 1362-1367.